Testing 3D games without SDK access

A global streaming platform offers cloud gaming, including a variety of casual games. Third parties submit the games, and each one must be verified before it is published. The platform provider wanted AI agents to perform autonomous testing, without SDK or codebase access, on released games to evaluate feasibility.

THE GOAL

Verify every game the way a real user plays it: on the TV or browser screen, steering from a phone, with AI agents doing the playing and no access to the games' internals.

Company
A global streaming platform

Industry
Cloud Gaming

Model
Games on a TV or browser window, played with phones as controllers

Testing types
Functional, Agent based

Pilot
Six weeks with 72 checks in scope

THE CHALLENGE AND HOW WE SOLVED IT

Aiming a shot with no way to measure it

A camera model turns the screenshot into a top-down grid, and a local model tests thousands of shots in seconds.

Reading a 3D course from a moving camera

A visual model (MobileSAM) outlines the course and walls, and image analysis finds the ball and the cup.

Working without code, SDK or documented controls

The agent plays through the screen and a phone controller, and the simulation shows what each action causes.

APPROACH AT A GLANCE

1. Agree what can be verified

We explained which checks could not be met as written and agreed a deliverable version.

2. Track everything

A tracker marks each check proven, possibly flaky or to do, and doubles as the status report.

3. Screenshot to simulation

The agent finds the ball, target and walls, then simulates thousands of shots.

4. Run the heavy work locally

A local model returns the best power and direction in under five seconds.

5. Recreate the course in 3D

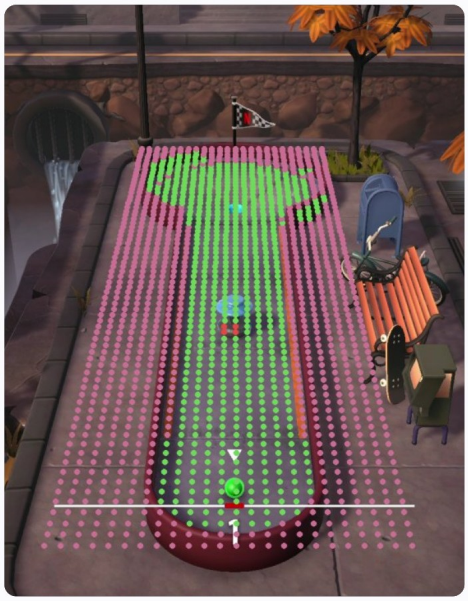
AI rebuilt the course as a Unity model, so simulations run in an accurate copy.

6. Orchestrate multiplayer

A supervisor agent coordinates player agents on remote machines.

24,000 shots from one screenshot

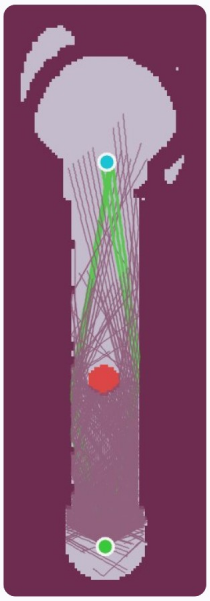
The agent finds the ball and the target, outlines the course, lays a grid of cells over it, simulates every shot at once, then picks the best one and takes it.



A grid of cells placed on the screen. Green cells fall inside the course outline.



The same roll put back on the screen: one bounce off the right rail, then into the cup.



242 sample shots; the green ones sink.

LOCAL MODEL

Agentic QA normally calls cloud models. Here a local model runs the simulations and returns the best power and direction.

3D COPY

AI rebuilt the course as a Unity model, so simulations run in an accurate 3D copy.

MULTIPLAYER

A supervisor agent coordinates player agents on remote machines, with actions landing within a few seconds.

24,000	5s or less	35	9 in 10
simulated shots per screenshot	to compute the best shots	holes sunk in live games	shots inside the 30-second timer

Median 17 seconds from turn start to the ball moving, across 85 live shots, Oct 5–7 2026. The planner reports 23,760 trajectories per search.

WHAT WORKED

A screenshot became the start of a simulation, and the heavy calculation moved to a local model.

WHERE IT APPLIES

Black-box testing of games with no SDK or source, and any case with several actors in different roles.

WHAT TRAVELS

The simulations are game-specific. The approach is not.